

Conditional Portfolio Decisions under Epistemic Uncertainty and Temporal Shift: Structured Value Sets and Calibrated No-Trade Gates

Tiantian Zhang

Department of Computer Science, Columbia University

t.zhang8@columbia.edu tiantianzhang02@bokesimu.com

Jierui Zuo

Department of Management Science and Engineering, Tsinghua University

zuojr22@mails.tsinghua.edu.cn

Kongchang Feng

Department of Physics, Columbia University

kf2824@columbia.edu

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Abstract

Machine-learning portfolio systems can fail for an operationally simple reason: they trade when the estimated active edge is not reliable enough to survive risk, costs, and temporal shift. We therefore treat equal weight or a robust optimization portfolio as the default and learned policies as bounded active overlays. The framework separates policy proposal from trade authorization. A conservative fitted-Q learner proposes a finite-action portfolio, while a calibrated lower-confidence gate decides whether the proposal may replace the baseline. Following cross-model uncertainty quantification, we distinguish within-model predictive dispersion from disagreement across a heterogeneous ensemble. Following epistemic robust offline reinforcement learning, we represent the complete action-value vector by an ellipsoidal uncertainty set that preserves covariance across actions and yields a closed-form worst-case active advantage. Because the next-period return vector reveals the one-step utility of every finite candidate portfolio, a purged split-conformal calibration can provide simultaneous one-step coverage and a selection-valid no-trade rule. A point-in-time shift gate based on state novelty, maximum mean discrepancy, residual shift, and cross-model disagreement determines when that calibration should be considered inapplicable.

Empirically, we retain the complete earlier evidence rather than replacing it with the public run. The proprietary evidence comprises a 12-roll method comparison, a previous 75-block full OOS experiment over 2019–2024, a conditional-scenario allocation diagnostic, and a non-comparable convex-optimization study. In the previous proprietary full OOS experiment, the conservative $\kappa = 0.20$ policy attained net NAV 1.675 versus 1.491–1.524 for four equal-weight variants, but its net Sharpe was lower, 0.681 versus 0.725–0.753, with average turnover 0.092. Those values are retained as historical evidence because the raw panel and complete run artifacts are not publicly redistributed.

We additionally report a fully public Yahoo Finance replication with 4,278 aligned return observations and 1,500 OOS decisions in 75 rolling blocks over 2019–2024. At $\kappa = 0.20$, the CNN-LSTM CQL-FQI proxy holds the previous allocation on 991 dates (66.1%), achieves net NAV 1.5667, Sharpe 0.5985, maximum drawdown -0.3253 , and average daily ℓ_1 turnover 0.1320. All four equal-weight variants achieve higher terminal NAV (1.6010–1.6287) and Sharpe (0.8410–0.8727). Thus the two full-OOS tracks differ in terminal-wealth ranking, but agree that the learned policy does not dominate equal weight on risk-adjusted performance and that aggressive thresholds overtrade. The full heterogeneous ensemble, ellipsoidal calibration, conformal authorization, and shift gate remain a deployment design rather than an empirically validated system.

1 Introduction

Portfolio allocation is a sequential decision problem. The portfolio held today determines both the next realized return and the cost of tomorrow’s rebalance. Yet many machine-learning pipelines remain static or two-stage: a predictor minimizes forecast error and a separate optimizer later converts its output into weights. Integrated prediction-optimization instead differentiates through the optimizer, while reinforcement learning (RL) directly optimizes a

sequential reward. Both approaches can improve decision alignment, but neither prevents a model from trading on an unreliable estimate.

This problem is especially acute in financial data. Rolling samples are short, realized rewards are noisy, policies are evaluated under temporal rather than independent sampling, and a small error in a cross-asset ranking can create large turnover. Equal weight is consequently more than a convenient benchmark. When the state contains little stable information about conditional cross-asset differences, equal weight diversifies without estimating a fragile ranking. The relevant learning problem is not unconditional replacement of equal weight, but *conditional decision learning*: identify states in which an active deviation has a credible lower-bound advantage after costs.

We use a baseline-plus-overlay representation,

$$w_t = (1 - \alpha_t)w_t^{\text{base}} + \alpha_t w_t^{\text{model}}, \quad 0 \leq \alpha_t \leq \alpha_{\max}, \quad (1)$$

where w_t^{base} is equal weight, minimum variance, risk parity, or robust mean–variance allocation. The production rule is conservative: $\alpha_t = 0$ unless a model proposal passes both a calibrated lower-confidence test and point-in-time shift diagnostics.

The central revision is to separate *proposal uncertainty* from *authorization*. A conservative offline-RL model may rank candidate actions through a pessimistic Bellman objective, but its Q-values are not automatically calibrated. We therefore add a structured uncertainty layer. A heterogeneous model ensemble produces a distribution over the complete action-value vector. Within-model dispersion measures predictive variability; between-model variation and a cross-model kernel discrepancy measure sensitivity to model choice. An ellipsoidal uncertainty set retains the covariance of action values, so that uncertainty in the contrast between an active action and the baseline depends on their correlation. A separate conformal step calibrates the radius using one-step portfolio utilities, which are observable for every finite candidate action after the next return vector arrives.

Contributions. The paper makes four claims of increasing strength.

1. We formalize equal weight as a default when the observable state contains no conditional cross-asset information after costs.
2. We transfer the structural idea of combining self-consistency with cross-model disagreement (Hamidieh et al., 2026) from generated outputs to distributions over portfolio value vectors. A simple lemma shows that the proposed bounded-kernel discrepancy is controlled by total variation, while also clarifying that it is a diagnostic rather than a confidence bound.
3. Inspired by epistemic robust offline RL (Chenreddy and Delage, 2026), we construct ellipsoidal action-value sets and derive a closed-form lower bound for each active contrast. A purged conformal calibration on observable one-step utilities yields a simultaneous, selection-valid no-trade guarantee under exchangeability.
4. We preserve the earlier short rolling comparison, proprietary full-OOS experiment, conditional-scenario diagnostic, and non-comparable convex study, and add a reproducible public Yahoo Finance replication with a committed data manifest, exact experiment commit, public action-construction code, action counts, and 1,500 OOS decisions. The proprietary and public full-OOS runs are reported as separate evidence tracks: their terminal-wealth rankings differ, but neither establishes risk-adjusted dominance over equal weight.

Scope. The theory below is a deployment specification. Five empirical tracks are kept distinct: a 12-roll proprietary method comparison, a previous 75-block proprietary full-OOS experiment, a public 75-block Yahoo replication, a proprietary conditional-scenario diagnostic, and a larger non-comparable Boke convex study. The first three use a single learned architecture at a time, a finite candidate set, a CQL-style penalty, and fixed thresholds $\kappa \in \{0, 0.10, 0.20\}$ where applicable. The public run uses one random seed and one 20-basis-point cost setting. None of the evidence tracks implements heterogeneous cross-model uncertainty, ellipsoidal robust policy training, conformal radius calibration, or the explicit shift gate. We state these distinctions throughout to avoid pooling non-comparable results or attributing unrun ablations, statistical significance, or calibration guarantees to the reported tables.

2 Related Work and Positioning

2.1 Robust portfolios and prediction-decision alignment

Markowitz optimization provides the basic return–risk language (Markowitz, 1952). Modern robust and multi-period formulations add forecast uncertainty, trading costs, constraints, and dynamic planning (Boyd et al., 2017, 2024; Johansson et al., 2024; Almgren and Chriss, 2001). Smart predict-then-optimize and integrated MVO train predictions through downstream decision quality rather than pointwise error (Elmachtoub and Grigas, 2022; Butler and Kwon, 2022; Deng et al., 2025). These methods address objective mismatch, but optimizer-in-the-loop learning can be statistically noisy and computationally expensive in short rolling windows.

2.2 Offline RL and structured epistemic uncertainty

Fitted Q iteration (FQI) is a classical batch-RL method (Ernst et al., 2005). Conservative Q-learning (CQL) penalizes unsupported action values and is therefore natural for offline market data (Kumar et al., 2020). Portfolio applications also use direct reinforcement, actor-critic methods, and ensemble systems (Moody and Saffell, 2001; Lillicrap et al., 2016; Fujimoto et al., 2018; Jiang et al., 2024; Yang et al., 2020). Their main operational risk is not only approximation error but extrapolation to actions and states weakly represented in the historical sample.

Chenreddy and Delage’s epistemic robust soft actor-critic (ERSAC) models uncertainty as a compact set of entire Q-value vectors rather than unrelated per-action intervals (Chenreddy and Delage, 2026). Box sets recover ensemble-min style pessimism, while convex-hull and ellipsoidal sets preserve dependence across actions. We use the same geometric principle for a finite-action portfolio gate. Our empirical CQL-FQI proxy is not an implementation of ERSAC; the ellipsoidal construction is added as theory and a future baseline.

2.3 Cross-model uncertainty and temporal shift

Self-consistency can be misleading when a model repeatedly produces the same wrong answer. Hamidieh et al. address this in language models by complementing within-model consistency with cross-model disagreement from a small, diverse ensemble (Hamidieh et al., 2026). We transfer the distinction, not the text-semantic estimator, to portfolio value distributions. A portfolio model can be internally stable while materially disagreeing with other plausible models about which action dominates.

Distribution-shift diagnosis further distinguishes familiar covariate shift, conditional outcome shift, and unseen target states (Cai et al., 2023). Distributionally robust optimization, variance regularization, online uncertainty quantification, temporal model selection, and temporal predictive inference motivate cautious decisions when the rolling deployment law changes (Duchi and Namkoong, 2021, 2019; Namkoong and Duchi, 2016; Su and Zhu, 2023; Han et al., 2024a,b; Capponi et al., 2025). Maximum mean discrepancy (MMD) provides a reproducible kernel two-sample statistic for the proposed state and residual checks (Gretton et al., 2012). Split conformal prediction supplies finite-sample marginal coverage under exchangeability (Lei et al., 2018); temporal drift requires additional caution and empirical coverage monitoring.

3 Problem Setup and the Default Policy

Let $r_{t+1} \in \mathbb{R}^N$ be next-period returns and let $w_t \in \Delta^N$ be the post-trade portfolio,

$$\Delta^N = \{w \in \mathbb{R}^N : w_i \geq 0, \mathbf{1}^\top w = 1\}. \quad (2)$$

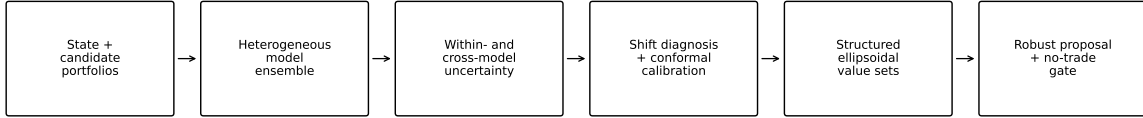
For proportional cost $c \geq 0$, a one-step risk-adjusted utility is

$$G_{t+1}(w) = w^\top r_{t+1} - c \|w - w_{t-1}\|_1 - \eta w^\top \Sigma_t^{\text{risk}} w, \quad (3)$$

where Σ_t^{risk} is known at decision time. The state is $s_t = (x_t, z_t, w_{t-1})$, with observed features x_t and learned representation $z_t = f_\theta(x_{t-L:t})$.

Table 1: How the principal literatures enter the revised framework.

Component	Representative work	Role here
Robust baseline	Markowitz; multi-period convex trading; robust MVO	Default portfolio, risk and turnover control.
Decision alignment	SPO/SPO+; integrated MVO; IPMO	Explains why forecast MSE need not imply decision quality.
Offline decision layer	FQI; CQL; portfolio RL	Proposes state-dependent actions while penalizing unsupported behavior.
Structured epistemic sets	ERSAC (Chenreddy and Delage, 2026)	Joint value-vector sets and covariance-aware robust contrasts.
Self/cross-model UQ	Hamidieh et al. (Hamidieh et al., 2026)	Flags model-choice uncertainty missed by within-model stability.
Calibration and shift	Conformal inference; DISDE; MMD; temporal inference	Calibrates the execution gate and determines when to abstain.



Execute a bounded active overlay only when the calibrated lower bound and every shift gate pass; otherwise use the baseline or hold.

Figure 1: Revised deployment architecture. Conservative RL proposes an action; structured uncertainty, calibration, and shift diagnostics authorize or reject it. The baseline is the output whenever any gate fails.

Let a_0 denote the baseline or hold action. A finite candidate set $\mathcal{A} = \{a_0, a_1, \dots, a_{K-1}\}$ maps each action to $w(a) \in \Delta^N$. A deployment set should additionally impose a bounded tilt,

$$\|w(a) - w(a_0)\|_1 \leq B, \quad a \in \mathcal{A}, \quad (4)$$

so that even an authorized decision cannot create an unconstrained portfolio jump.

Proposition 1 (Equal weight under absent conditional cross-asset information). *Let $w^{\text{EW}} = N^{-1}\mathbf{1}$ be the portfolio already held before the decision, and let δw_t be measurable with respect to X_t , with $\mathbf{1}^\top \delta w_t = 0$. Suppose*

$$\mathbb{E}[r_{t+1} \mid X_t] = m_t \mathbf{1}. \quad (5)$$

If an active deviation incurs a nonnegative incremental cost $C_t(\delta w_t) \geq 0$ relative to holding equal weight, then

$$\mathbb{E}[\delta w_t^\top r_{t+1} - C_t(\delta w_t)] \leq 0. \quad (6)$$

Proof. Conditioning on X_t , $\mathbb{E}[\delta w_t^\top r_{t+1} \mid X_t] = m_t \delta w_t^\top \mathbf{1} = 0$. Subtracting $C_t(\delta w_t) \geq 0$ and taking expectations proves the claim. $\square$

The proposition is intentionally narrow. It does not assert universal optimality of equal weight. It states the burden of proof for an active strategy; the state must contain stable information about cross-sectional returns or dependence, and that information must survive risk, estimation error, and turnover.

4 Structured Uncertainty and a Calibrated No-Trade Gate

4.1 A heterogeneous ensemble over action-value vectors

For each model $m \in \{1, \dots, M\}$ and state s_t , let

$$q_{t,m}^{(b)} = (Q_{t,m}^{(b)}(a_0), \dots, Q_{t,m}^{(b)}(a_{K-1}))^\top \in \mathbb{R}^K, \quad b = 1, \dots, B_q, \quad (7)$$

be predictive draws of the complete risk- and cost-adjusted action-value vector. Draws may come from residual bootstrap, posterior sampling, dropout, or scenario simulation. Let $\pi_m \geq 0$ with $\sum_m \pi_m = 1$ be fixed using only past validation data. Define

$$\hat{\mu}_{t,m} = \frac{1}{B_q} \sum_b q_{t,m}^{(b)}, \quad \hat{\Sigma}_{t,m}^{\text{within}} = \text{Cov}_b(q_{t,m}^{(b)}), \quad (8)$$

$$\hat{\mu}_t = \sum_m \pi_m \hat{\mu}_{t,m}, \quad \hat{\Sigma}_t^{\text{within}} = \sum_m \pi_m \hat{\Sigma}_{t,m}^{\text{within}}, \quad (9)$$

$$\hat{\Sigma}_t^{\text{between}} = \sum_m \pi_m (\hat{\mu}_{t,m} - \hat{\mu}_t)(\hat{\mu}_{t,m} - \hat{\mu}_t)^\top. \quad (10)$$

The total-covariance estimator is

$$\hat{\Sigma}_t^Q = \hat{\Sigma}_t^{\text{within}} + \hat{\Sigma}_t^{\text{between}}. \quad (11)$$

The first term is a within-model predictive-dispersion proxy; the second measures sensitivity to model choice. An effective ensemble should have support richness, non-collapsing diversity, and weights calibrated on past data. Identical copies of one misspecified model cannot reveal epistemic disagreement.

Within-model covariance can still be small when a model is confidently wrong. To complement it, let $P_{t,m}$ be model m 's predictive distribution over q and let $P_{t,-m} = \sum_{\ell \neq m} \pi_\ell P_{t,\ell}$ be the normalized leave-one-model-out mixture. For a positive-semidefinite kernel $k : \mathbb{R}^K \times \mathbb{R}^K \rightarrow [0, 1]$, define the self-versus-cross discrepancy

$$D_{t,m} = \mathbb{E}_{q,q' \sim P_{t,m}} k(q, q') - \mathbb{E}_{q \sim P_{t,m}, \tilde{q} \sim P_{t,-m}} k(q, \tilde{q}), \quad D_t^{\text{cross}} = \sum_m \pi_m [D_{t,m}]_+. \quad (12)$$

An RBF kernel on standardized value vectors gives a direct empirical estimator by replacing expectations with averages over predictive draws. Equation (12) is the vector-valued analogue of comparing intra-model and inter-model consistency (Hamidieh et al., 2026).

Lemma 2 (The cross-model score is controlled by distributional mismatch). *For $0 \leq k \leq 1$,*

$$|D_{t,m}| \leq \text{TV}(P_{t,m}, P_{t,-m}). \quad (13)$$

Proof. Let $f(y) = \mathbb{E}_{q \sim P_{t,m}} k(q, y)$. Then $0 \leq f \leq 1$ and $D_{t,m} = \mathbb{E}_{P_{t,m}} f - \mathbb{E}_{P_{t,-m}} f$. The variational definition of total variation over functions bounded in $[0, 1]$ gives the result. $\square$

Remark 3 (Diagnostic, not confidence radius). D_t^{cross} is neither symmetric nor, by itself, a calibrated upper bound on value error. It is used as a model-mismatch and abstention diagnostic. The radius of the lower-confidence set is calibrated separately. This distinction avoids treating generic model disagreement as a guarantee.

4.2 Ellipsoidal value sets and covariance-aware active advantages

Let $\varepsilon_t^Q = \max\{10^{-12}, 10^{-6} \text{tr}(\hat{\Sigma}_t^Q)/K\}$ regularize the covariance and define

$$\mathcal{U}_t^Q(\tau) = \left\{ q \in \mathbb{R}^K : (q - \hat{\mu}_t)^\top (\hat{\Sigma}_t^Q + \varepsilon_t^Q I)^{-1} (q - \hat{\mu}_t) \leq \tau^2 \right\}. \quad (14)$$

This is the finite-action ellipsoidal construction motivated by ERSAC (Chenreddy and Delage, 2026). Unlike coordinate-wise lower bounds, it retains inter-action covariance. Let e_a be the coordinate vector for action a and define the active contrast $d_a = e_a - e_{a_0}$.

Proposition 4 (Closed-form worst-case active value). *For any action a ,*

$$\inf_{q \in \mathcal{U}_t^Q(\tau)} d_a^\top q = d_a^\top \hat{\mu}_t - \tau \sqrt{d_a^\top (\hat{\Sigma}_t^Q + \varepsilon_t^Q I) d_a}. \quad (15)$$

Proof. Write $q = \hat{\mu}_t + (\hat{\Sigma}_t^Q + \varepsilon_t^Q I)^{1/2} u$ with $\|u\|_2 \leq \tau$. The minimum of the linear term $d_a^\top (\hat{\Sigma}_t^Q + \varepsilon_t^Q I)^{1/2} u$ over the Euclidean ball is minus τ times its norm, yielding Equation (15). $\square$

The uncertainty term expands as

$$d_a^\top \hat{\Sigma}_t^Q d_a = \hat{\Sigma}_{aa}^Q + \hat{\Sigma}_{a_0 a_0}^Q - 2\hat{\Sigma}_{aa_0}^Q. \quad (16)$$

Thus two individually uncertain actions may have a precise difference when their errors move together, while independent error bars would miss this cancellation. A robust proposal is

$$a_t^{\text{prop}} \in \arg \max_{a \in \mathcal{A}} \left\{ d_a^\top \hat{\mu}_t - \tau_Q \sqrt{d_a^\top (\hat{\Sigma}_t^Q + \varepsilon_t^Q I) d_a} \right\}. \quad (17)$$

This is a deterministic-action specialization of robust optimization over a joint value-vector set. It is a policy proposal, not yet permission to trade. The proposal radius τ_Q is selected on a pre-deployment tuning block from a fixed grid; unlike the conformal radius below, it carries no coverage claim.

4.3 Observable calibration of the execution gate

Direct calibration of a long-horizon optimal Q-function is difficult because the counterfactual continuation value is not observed. The finite-action portfolio setting nevertheless provides a useful full-information object. After r_{t+1} is observed, Equation (3) can be evaluated for every candidate weight, not only for the action actually taken. Define

$$G_{t+1} = (G_{t+1}(w(a_0)), \dots, G_{t+1}(w(a_{K-1})))^\top. \quad (18)$$

A model ensemble produces a one-step center $\hat{\mu}_t^G$ and shape matrix $\hat{\Sigma}_t^G$ by the same within/between decomposition as Equation (11).

Use three chronological, purged blocks: a fit block, a gate-tuning block, and a conformal block. A reference implementation assigns 60%, 20%, and 20% of the available history before inserting an H -period purge between adjacent blocks. Gate thresholds are frozen on the gate-tuning block. The conformal set $\mathcal{C} = \{j_1, \dots, j_n\}$ contains only historical pseudo-deployment points accepted by those same frozen thresholds. Predictions for $j \in \mathcal{C}$ must be generated without using G_{j+1} . Let $\varepsilon_j^G = \max\{10^{-12}, 10^{-6} \text{tr}(\hat{\Sigma}_j^G)/K\}$. Define Mahalanobis nonconformity scores

$$S_j = \sqrt{(G_{j+1} - \hat{\mu}_j^G)^\top (\hat{\Sigma}_j^G + \varepsilon_j^G I)^{-1} (G_{j+1} - \hat{\mu}_j^G)}. \quad (19)$$

For target miscoverage δ , let

$$k_\delta = \lceil (n+1)(1-\delta) \rceil, \quad \hat{\tau}_\delta = S_{(k_\delta)}, \quad (20)$$

where $S_{(k)}$ is the k th order statistic and $S_{(n+1)} := +\infty$.

Assumption 5 (Local exchangeability for one-step calibration). *Conditional on the frozen deployment gate accepting the current regime, the accepted conformal examples and the next pair (s_t, G_{t+1}) are exchangeable with respect to the score in Equation (19).*

Proposition 6 (Simultaneous one-step coverage). *Under local exchangeability and the purged split construction,*

$$\Pr \{G_{t+1} \in \mathcal{U}_t^G(\hat{\tau}_\delta)\} \geq 1 - \delta, \quad (21)$$

where $\mathcal{U}_t^G$ is defined as in Equation (14) with $(\hat{\mu}_t^G, \hat{\Sigma}_t^G)$ and $\varepsilon_t^G = \max\{10^{-12}, 10^{-6} \text{tr}(\hat{\Sigma}_t^G)/K\}$.

Proof. The n calibration scores and the next score are exchangeable. The rank of the next score is therefore uniform over $\{1, \dots, n+1\}$ up to ties. The conformal order statistic in Equation (20) gives coverage at least $1 - \delta$. $\square$

Define the calibrated lower bound on realized one-step active utility,

$$\text{LCB}_t^G(a) = d_a^\top \hat{\mu}_t^G - \hat{\tau}_\delta \sqrt{d_a^\top (\hat{\Sigma}_t^G + \varepsilon_t^G I) d_a}. \quad (22)$$

Corollary 7 (Selection-valid no-trade gate). *Suppose the policy executes a data-dependent proposal a_t^{prop} only when $\text{LCB}_t^G(a_t^{\text{prop}}) > \kappa$. On the coverage event in Equation (21),*

$$G_{t+1}(w(a_t^{\text{prop}})) - G_{t+1}(w(a_0)) > \kappa. \quad (23)$$

Consequently the executed action has realized one-step active utility greater than κ with probability at least $1 - \delta$.

Proof. Vector coverage implies every linear contrast is bounded below by the ellipsoid support function, including the adaptively selected contrast $d_{a_t^{\text{prop}}}$. Equation (22) and the switching condition complete the proof. $\square$

This result is stronger and more testable than assuming an unspecified error bar $U_t(a)$. It is also deliberately limited: it guarantees one-step realized utility under the stated exchangeability condition, not a time-uniform lower bound on the unobserved optimal Q-function. Extending the result to multi-step value requires valid off-policy confidence sets or calibrated Bellman residual propagation.

4.4 A concrete point-in-time shift gate

Global exchangeability is implausible in markets. We therefore specify a reproducible abstention rule rather than an unspecified mapping $g(\cdot)$. All standardization, kernels, bandwidths, and thresholds are computed from data available before time t .

Let Z^{fit} be standardized fit-block representations, Z_t^{recent} a trailing 20-trading-day deployment window, and $E_j = G_{j+1} - \hat{\mu}_j^G$ realized calibration residual vectors. Use an RBF kernel whose bandwidth is the median pairwise fit-block distance and a five-nearest-neighbor novelty score. Construct four scores:

$$U_t^{\text{novel}} = k\text{NN distance from } z_t \text{ to } Z^{\text{fit}}, \quad (24)$$

$$U_t^{\text{fam}} = \text{MMD}^2(Z^{\text{fit}}, Z_t^{\text{recent}}), \quad (25)$$

$$U_t^{\text{cond}} = \text{MMD}^2(E^{\text{cal}}, E_t^{\text{recent}}), \quad (26)$$

$$U_t^{\text{model}} = D_t^{\text{cross}}. \quad (27)$$

The first score identifies unseen states; the second measures familiar covariate drift; the third is a lagged proxy for change in the conditional payoff law after accounting for the predictor; the fourth detects model-specific disagreement. Generate null score distributions by pseudo-deployment splits inside the gate-tuning data and set each threshold to its predeclared $(1 - \xi)$ quantile. The reference defaults are $\xi = 0.05$ and conformal miscoverage $\delta = 0.10$; both must be fixed before OOS evaluation and included in sensitivity analysis. Then

$$\chi_t = \mathbf{1} \{U_t^{\text{novel}} \leq q_{\text{novel}}, U_t^{\text{fam}} \leq q_{\text{fam}}, U_t^{\text{cond}} \leq q_{\text{cond}}, U_t^{\text{model}} \leq q_{\text{model}}\}. \quad (28)$$

The complete decision is

$$a_t = \begin{cases} a_t^{\text{prop}}, & \chi_t = 1 \text{ and } \text{LCB}_t^G(a_t^{\text{prop}}) > \kappa, \\ a_0, & \text{otherwise.} \end{cases} \quad (29)$$

A hard gate is intentionally used as the default. A soft mapping from scores to α_t may be tuned later, but it should not replace the auditable abstention rule. The gate does not magically restore exchangeability; it narrows deployment to states resembling the calibration regime. Empirical coverage, rejection frequency, and performance conditional on acceptance must be reported.

Table 2: Implementation status. A checkmark means that the component appears in at least one reported seven-ETF evidence track; it does not imply validation of the complete calibrated gate.

Component	Reported evidence	Calibrated deployment design
Equal-weight/hold default and explicit costs	✓	✓
Finite portfolio candidate set	✓	✓
CQL-style penalty and fixed switch threshold	✓	✓
Previous proprietary 75-block full-OOS experiment	✓	historical evidence
Conditional-scenario and larger convex diagnostics	✓	supporting evidence
Public Yahoo panel, manifest, code SHA, and OOS outputs	✓	reproducibility layer
Public action-construction code and threshold action counts	✓	reproducibility layer
Heterogeneous cross-model ensemble	×	✓
Within/between covariance decomposition	×	✓
Ellipsoidal joint Q-value set	×	✓
Purged conformal radius calibration	×	✓
Explicit novelty/MMD/residual/model shift gate	×	✓
Same-universe risk-parity/min-var/robust-MVO results	×	required
Seed, cost, window sensitivity and block-bootstrap intervals	×	required

4.5 Relation to conservative FQI

For a transition (s_t, a_t, R_t, s_{t+1}) , the retained empirical proxy used

$$y_t = R_t + \gamma \max_{a'} Q_{\bar{\psi}}(s_{t+1}, a'), \quad \mathcal{L}_Q = (Q_{\psi}(s_t, a_t) - y_t)^2, \quad (30)$$

with the CQL-style penalty

$$\mathcal{L}_{\text{CQL}} = \log \sum_{a \in \mathcal{A}} \exp Q_{\psi}(s_t, a) - Q_{\psi}(s_t, a_{\text{hold/base}}). \quad (31)$$

An auxiliary return-prediction head regularizes the latent representation. CQL pessimism and the calibrated execution gate solve different problems: CQL reduces extrapolation error during value learning; Equations (22)–(29) determine whether a proposed action is sufficiently credible to deploy.

5 Empirical Diagnostics and Public Replication

5.1 Data, rolling design, and provenance

The evidence has five distinct tracks. First, a short internal Wind/Boke diagnostic contains 12 rolling blocks and 240 daily decisions. Second, the prior manuscript reports a proprietary seven-ETF full-OOS experiment with 75 rolling blocks and 1,500 decisions over 2019–2024, using a 250-day lookback, 120-day state window, 20-day test blocks, and 20 basis points times ℓ_1 turnover. Its tables and curves are retained exactly as previously reported, but the raw panel and complete execution artifacts are unavailable for public audit. Third, the public replication uses adjusted Yahoo Finance prices for AGG, DBC, GLD, IWM, LQD, MUB, and VTI, mapped to the repository labels AGG.P, DBC.P, GLD.P, IWM.P, LQD.P, MUB.P, and VTI.P. Prices span 2008-01-02 through 2024-12-31; after intersecting observed trading dates without price or return imputation, the return panel contains 4,278 observations from 2008-01-03 through 2024-12-31. Its OOS period is 2019-01-03 through 2024-12-17, again comprising 75 rolling blocks and 1,500 daily decisions. Fourth, a separate proprietary conditional-scenario diagnostic studies whether state conditioning matters more than generator choice. Fifth, the appendix retains a larger Boke equity/statistical-arbitrage convex comparison that is not numerically comparable to either seven-ETF panel.

The public run uses a 250-day training lookback, a 120-day state sequence, 20-day test blocks, one fitted-Q iteration, one epoch per iteration, hidden size 16, and seed 2021. Proportional costs are 20 basis points times daily ℓ_1 turnover in

Table 3: Empirical evidence retained in the manuscript. Tracks with different data provenance are reported separately.

Track	Data and scale	Main comparison	Evidentiary role
Short internal rolling	Proprietary seven ETFs; 12 blocks, 240 decisions	Two-stage, integrated, FQI, CQL-FQI, EW	Method and turnover diagnostic
Previous proprietary full OOS	Proprietary seven ETFs; 75 blocks, 1,500 decisions	Four EW variants and three thresholds	Historical full-horizon evidence
Public Yahoo full OOS	Public seven ETFs; 75 blocks, 1,500 decisions	Same EW family and thresholds	Reproducible public replication
Conditional scenarios	Proprietary seven ETFs; 2019–2024	Unconditional, regime, and state-conditioned scenarios	Conditioning-versus-generator diagnostic
Boke convex study	Larger proprietary equity/stat-arb setting	Robust MVO and related optimizers	Non-comparable provenance

both training rewards and evaluation. The exact source checkout is [f492551](#) on the [public Yahoo replication branch](#); the manifest and archived summary outputs are committed with the run.¹

All reported comparisons remain descriptive. The proprietary full-OOS and public Yahoo runs share a nominal date range and rolling scale but not a redistributable common input panel or a common archived execution environment; their point estimates must not be pooled or interpreted as an out-of-sample replication pair. The public replication has a single seed and a single cost setting, and it does not provide pre-registered confidence intervals. The sweep over κ is a threshold sensitivity diagnostic, not a substitute for cost, seed, training-window, or statistical-significance analysis.

5.2 Compared methods and finite actions

The short internal diagnostic compares equal weight, two-stage prediction plus downstream optimization, integrated optimizer-in-the-loop learning, plain FQI, and Deep CQL-FQI. The previous proprietary full-OOS experiment and the public replication each compare four equal-weight conventions with a CNN–LSTM CQL-FQI proxy at $\kappa \in \{0, 0.10, 0.20\}$, but they are distinct runs with different data provenance. The public action library contains 11 date-specific choices: hold the previous portfolio, equal weight, inverse volatility, inverse variance, bond and equity baskets, 100% gold, 100% commodity, top-two and top-three risk-adjusted momentum portfolios, and a softmax momentum portfolio. Volatility uses 20 days, momentum uses 60 days, and the first 120 observations use equal-weight warmup actions. Appendix B gives the complete construction and the archived selection counts. Thus the public action vectors are dynamic rather than a single fixed numerical grid, but their formulas and static basket weights are reproducible from public code. The prior proprietary source described the same action families at a higher level but did not preserve an auditable date-by-date weight tensor.

5.3 Short internal rolling comparison

Equal weight is strongest. The high-threshold CQL-FQI proxy improves over the learned two-stage, integrated, and plain-FQI variants, but still does not deliver positive active value over equal weight. The $\kappa = 0$ variant has plausible gross Sharpe and substantially worse net Sharpe, illustrating the central role of no-trade discipline.

5.4 Previous proprietary 2019–2024 full-OOS experiment

The earlier manuscript also reported a longer proprietary-panel experiment with 75 rolling blocks and 1,500 OOS decisions. We restore it here because it is part of the original empirical record. The values below are transcribed from the supplied prior source and the vector curves are the original figures. They are not presented as a public reproduction,

¹The public tables use the archival bundle under `artifacts/public_yahoo_colab_f492551/outputs/full_oos` and the accompanying [public OOS report](#). The historical proprietary tables are sourced instead from the prior manuscript archive supplied by the authors.

Table 4: Internal 12-roll seven-ETF diagnostic on the proprietary panel. Equal weight remains the strongest benchmark.

Method	Net NAV	Net Sharpe	Gross Sharpe	Avg. turnover
Equal weight	1.144	2.769	2.769	0.000
Deep CQL-FQI, $\kappa = 0.20$	1.108	1.690	1.848	0.021
Two-stage, $H = 10$	1.074	1.477	—	0.123
Deep CQL-FQI, $\kappa = 0.10$	1.097	1.397	2.001	0.089
Deep CQL-FQI, $\kappa = 0.00$	1.099	1.291	2.715	0.212
Integrated, $H = 10$	1.057	1.145	—	0.155
Plain FQI, $\gamma = 0$	1.082	0.697	1.867	0.297
Plain FQI, $\gamma = 0.95$	0.995	0.030	0.998	0.259

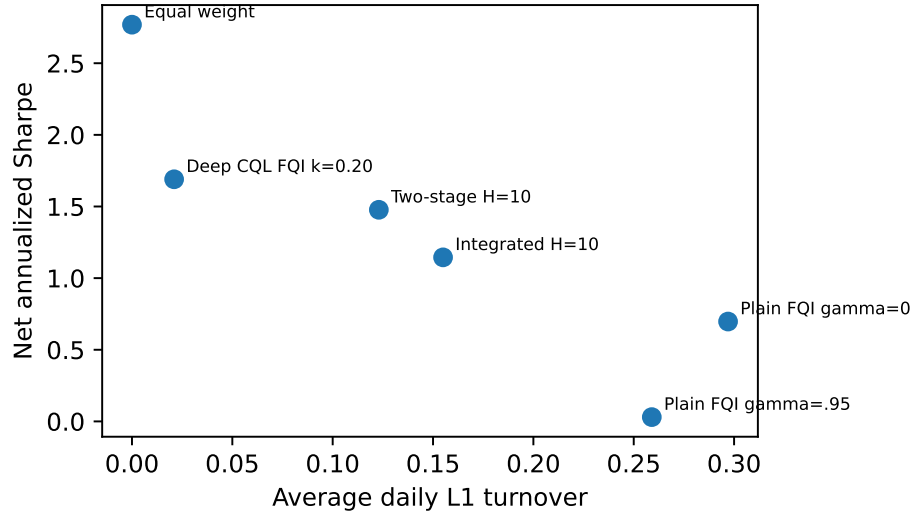


Figure 2: Turnover versus net Sharpe in the short internal diagnostic. Lowering the fixed threshold raises turnover and sharply reduces net performance.

and no confidence intervals or complete run metadata are available. In particular, the matching date span and nominal block count do not make this experiment numerically interchangeable with the Yahoo run in Section 5.5.

The historical run provides a different terminal-wealth ordering from the public Yahoo replication. Nevertheless, its substantive conclusion is already conservative: the higher terminal NAV is accompanied by lower Sharpe, nontrivial turnover, and only small active information ratios, while the lower thresholds lose heavily. The defensible cross-track statement is therefore not that the proprietary result has been confirmed publicly. It is that the high threshold repeatedly suppresses the worst overtrading behavior, whereas evidence for robust active value remains sensitive to data provenance and evaluation details.

Table 5: Previously reported proprietary seven-ETF full-OOS experiment over 2019–2024. The conservative learned policy has the highest terminal NAV but lower Sharpe than every equal-weight convention. Values are retained at the precision of the prior manuscript.

Method	Net NAV	Net Sharpe	Max DD	Avg. turnover
EW zero-cost target	1.517	0.753	-0.207	0.000
EW monthly rebalanced	1.495	0.734	-0.207	0.001
EW buy-and-hold	1.524	0.732	-0.208	0.000
EW daily rebalanced	1.491	0.725	-0.208	0.006
CNN-LSTM CQL-FQI, $\kappa = 0.20$	1.675	0.681	-0.203	0.092
CNN-LSTM CQL-FQI, $\kappa = 0.10$	0.686	-0.248	-0.415	0.199
CNN-LSTM CQL-FQI, $\kappa = 0.00$	0.589	-0.432	-0.468	0.257

Table 6: Previously reported active metrics for the proprietary $\kappa = 0.20$ policy. Active NAV is positive, but information ratios are small and active drawdowns are material.

Baseline	Active NAV	Active IR	Hit rate	Active max DD
EW buy-and-hold	1.097	0.206	0.489	-0.233
EW daily rebalanced	1.123	0.244	0.516	-0.218
EW monthly rebalanced	1.122	0.243	0.493	-0.220
EW zero-cost target	1.104	0.216	0.491	-0.222

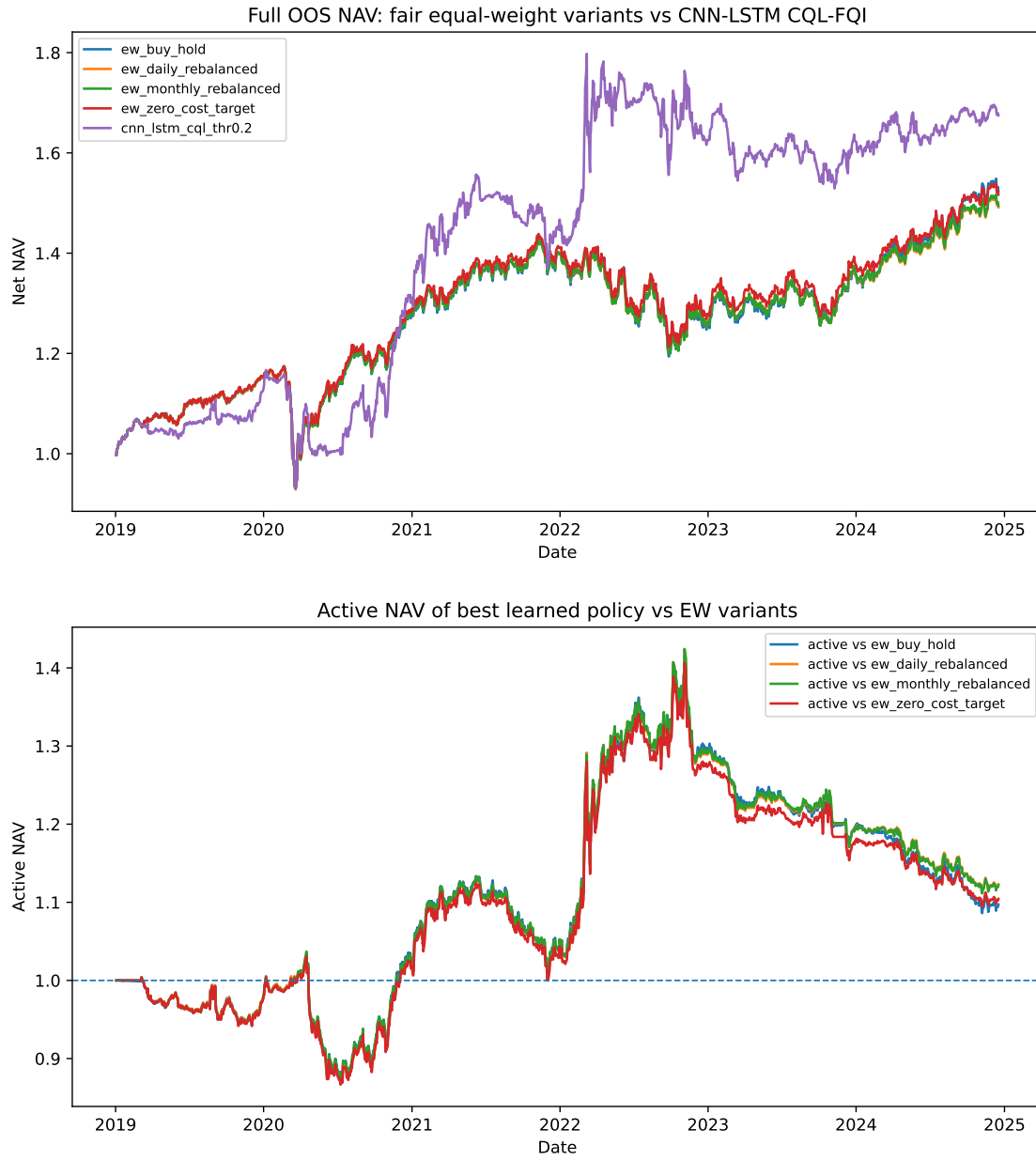


Figure 3: Original curves for the previous proprietary full-OOS experiment. Top: total NAV; the $\kappa = 0.20$ policy ends above the equal-weight variants but follows a more volatile path. Bottom: active NAV relative to each equal-weight convention; the cumulative edge is positive but reverses materially after its peak.

5.5 Public Yahoo 2019–2024 replication

The threshold sweep supports a limited, asymmetric conclusion. Moving from $\kappa = 0$ to $\kappa = 0.20$ raises the hold rate from 24.1% to 66.1%, lowers average turnover from 0.3253 to 0.1320, and avoids the severe losses of the aggressive policies. Costs remain material: the $\kappa = 0.20$ policy’s gross Sharpe is 1.0695, compared with a net Sharpe of 0.5985. Nevertheless, its terminal NAV, Sharpe, and drawdown are all worse than the equal-weight alternatives, and its active NAV is below one against each. Unlike the previous proprietary full-OOS experiment, the public run does not show a terminal-wealth advantage. Because the inputs and execution provenance differ, this divergence should be treated as evidence of fragility rather than used to select one preferred point estimate. The public result corroborates the warning about overtrading, not a claim of learned alpha or superiority over passive diversification.

Table 7: Public Yahoo Finance OOS point estimates over 1,500 daily decisions. Every equal-weight convention exceeds every learned threshold in both terminal NAV and net Sharpe.

Method	Net NAV	Net Sharpe	Max DD	Avg. turnover
EW zero-cost target	1.6280	0.8727	-0.2065	0.0000
EW monthly rebalanced	1.6034	0.8537	-0.2069	0.0012
EW daily rebalanced	1.6010	0.8444	-0.2072	0.0056
EW buy-and-hold	1.6287	0.8410	-0.2070	0.0000
CNN-LSTM CQL-FQI, $\kappa = 0.20$	1.5667	0.5985	-0.3253	0.1320
CNN-LSTM CQL-FQI, $\kappa = 0.10$	0.5856	-0.4423	-0.4985	0.2390
CNN-LSTM CQL-FQI, $\kappa = 0.00$	0.4459	-0.6828	-0.6135	0.3253

Table 8: Abstention and performance across fixed thresholds in the public run. “Hold” is the action that leaves the previous allocation unchanged.

κ	Hold count	Hold rate	Active dates	Avg. turnover	Net Sharpe
0.00	361	24.1%	1,139	0.3253	-0.6828
0.10	604	40.3%	896	0.2390	-0.4423
0.20	991	66.1%	509	0.1320	0.5985

Table 9: Active point estimates for the public $\kappa = 0.20$ policy. Active NAV is below one against every equal-weight convention, and active information ratios are economically close to zero.

Baseline	Active NAV	Active IR	Hit rate	Active max DD
EW buy-and-hold	0.9620	-0.0168	0.4773	-0.3067
EW daily rebalanced	0.9809	0.0170	0.5633	-0.3037
EW monthly rebalanced	0.9804	0.0154	0.4900	-0.3025
EW zero-cost target	0.9646	-0.0114	0.4127	-0.3077

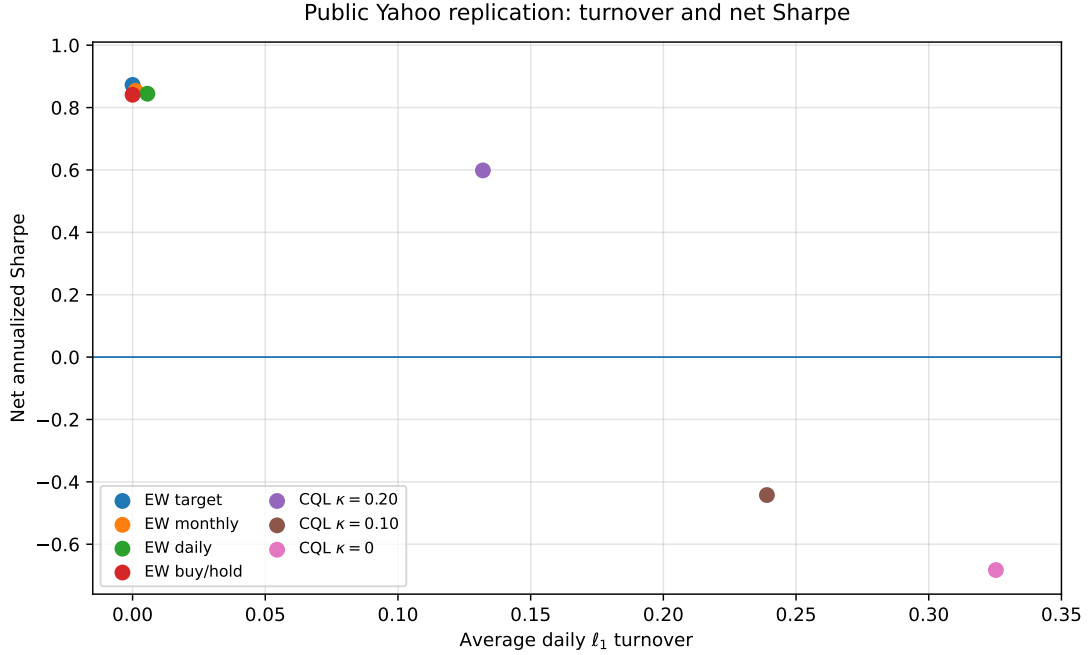


Figure 4: Public Yahoo replication. A larger fixed threshold increases abstention and sharply reduces turnover, but even $\kappa = 0.20$ remains below all equal-weight variants in net Sharpe.

5.6 Conditional scenario learning as supporting evidence

A separate proprietary-data, CPU-only diagnostic asks whether active value comes from conditioning on the joint ETF state rather than from the RL architecture itself. It is not a rerun on the public Yahoo panel and should not be numerically compared with Table 7. Models are fitted before 2019 and evaluated through 2024. At each decision date they construct joint 20-day scenarios and allocate a portfolio constrained to a 25% active tilt around equal weight, with the same 20-basis-point turnover charge.

Table 10: Conditional-scenario diagnostic over 2019–2024. State conditioning is associated with the main improvement; the incremental difference between conditional Gaussian and flow-based scenarios is modest.

Method	Net NAV	Sharpe	Max drawdown
Historical bootstrap	1.4472	0.6576	-0.2243
Global Gaussian	1.4888	0.6950	-0.2076
Equal weight	1.5268	0.7644	-0.2070
Regime bootstrap	1.5470	0.7896	-0.2045
Conditional Gaussian	1.5879	0.8392	-0.1994
Conditional flow-based scenarios	1.5949	0.8551	-0.1972

Unconditional historical and Gaussian scenarios remain below equal weight. Regime conditioning moves above it, and full state conditioning produces the largest gain. The Sharpe difference between the conditional flow-based and conditional Gaussian models is only 0.0159. This pattern is consistent with the paper’s interpretation that state information is more important than selecting a particular flexible generator. It remains a single-universe diagnostic without formal uncertainty intervals.

5.7 Baseline and comparator coverage

The public replication closes the public-data and exact-provenance gap for equal weight and fixed-threshold CQL-FQI, but it does not close the comparator gap. Risk parity, minimum variance, maximum Sharpe, and robust MVO have not been run on the same public panel under the same cost convention. Modern offline-RL comparators such as IQL, BCQ/BRAC variants, SPIBB, and ERSAC, as well as Bayesian or distributional dynamic programming, are also absent. Table 11 separates public results from internal-only diagnostics.

Table 11: Comparator scope across the public and internal seven-ETF evidence.

Comparator family	Public Yahoo	Internal diagnostic	Same 20 bp convention
Equal-weight variants	✓	✓	✓
CQL-FQI fixed thresholds	✓	✓	✓
Two-stage / integrated / plain FQI	×	✓	✓
Risk parity / minimum variance / maximum Sharpe	×	×	specified only
Robust MVO	×	non-comparable appendix only	specified only
IQL / BCQ / BRAC / SPIBB / ERSAC	×	×	—
Bayesian or distributional dynamic programming	×	×	—

For reference, Appendix A records the optimization definitions and a non-comparable internal Boke table. Those values use a larger equity/statistical-arbitrage setting and are not evidence that the learned overlay beats convex baselines on the seven ETFs.

6 Discussion

Why the theory is a gate rather than another return model. The empirical prototype already contains a CQL penalty and a fixed switch threshold. The new theory does not claim a new alpha source. It specifies how multiple plausible models, action covariance, observable calibration, and shift diagnostics can convert a heuristic threshold into an auditable abstention rule. This is valuable precisely because the existing active signal is modest.

Why cross-model disagreement and ellipsoids are complementary. Between-model covariance and the kernel discrepancy address model-choice sensitivity. The ellipsoid then converts the joint covariance into action-specific contrast uncertainty. Neither component alone gives coverage: disagreement can be uncalibrated, while a calibrated radius built from collapsed models may be too narrow in new regimes. The explicit shift gate and empirical coverage report are therefore part of the method, not optional diagnostics.

Why one-step calibration is both useful and limited. The finite action set makes every next-period candidate payoff observable ex post, which supports a clean simultaneous conformal statement. This is an unusual advantage of portfolio allocation over bandit feedback. The guarantee does not certify long-horizon Q-values. A practical system can therefore use robust Q-values to rank actions and a calibrated one-step utility bound to authorize execution, while treating multi-step uncertainty as a separate research problem.

Why equal weight remains the null hypothesis. The complete evidence is mixed in terminal wealth but consistent in risk-adjusted interpretation. The previous proprietary full-OOS run places the $\kappa = 0.20$ policy above equal weight in terminal NAV but below every equal-weight convention in Sharpe; the public run places it below equal weight in both. Both show severe deterioration at lower thresholds. Since the two long runs do not share a publicly auditable input panel and execution archive, their difference is itself a warning about dataset and pipeline sensitivity. The appropriate conclusion is not that the overlay dominates, but that a high threshold prevents the most damaging overtrading. Any

deployment should continue to default to a passive or robust convex portfolio unless a calibrated lower bound establishes a credible state-conditional edge.

7 Limitations and Reproducibility Requirements

The revision resolves one important weakness by providing a public seven-ETF panel, an executable data workflow, a committed manifest, public action-construction code, an exact experiment SHA, and archived OOS summaries, while also restoring the earlier experiments. Eight material limitations remain. First, seven broad ETFs still form a small universe favorable to passive diversification. Second, the public result is a single-seed run at one 20-basis-point cost and one training-window configuration. Third, no block-bootstrap confidence intervals, formal performance tests, or regime-stratified uncertainty analysis are reported. Fourth, same-universe risk-parity, minimum-variance, maximum-Sharpe, and robust-MVO comparators remain absent. Fifth, modern safe/offline-RL baselines and off-policy-evaluation checks remain absent. Sixth, although the public formulas reconstruct the date-varying action vectors, a date-by-date action-weight tensor is not separately archived with the summary tables. Seventh, the heterogeneous ensemble, cross-model score, ellipsoidal value set, conformal calibration, and explicit shift gate have not yet been run on any evidence track. Eighth, the prior full-OOS, short rolling, conditional-scenario, and Boke convex experiments remain tied to proprietary data and lack the complete artifacts needed for independent reproduction; their inclusion improves completeness, not auditability.

A confirmatory version should therefore pre-register and release: the full date-level action tensor and OOS return paths; fit, calibration, and test dates; all architecture and optimizer settings; at least five seeds; costs at 5, 10, 20, and 50 basis points; multiple training-window lengths; block-bootstrap intervals for Sharpe, active IR, and terminal-NAV differences; empirical coverage and reliability plots for Equation (22); gate acceptance rates by market regime; covariance-shrinkage and kernel-bandwidth ablations; and same-universe passive, convex, and modern offline-RL comparators. These are validation requirements, not claims about completed experiments.

8 Conclusion

This paper reframes dynamic portfolio learning as a conditional authorization problem under epistemic uncertainty and temporal shift. Equal weight or a robust optimization portfolio is the default. Conservative RL proposes a bounded action, a heterogeneous ensemble represents uncertainty over the joint action-value vector, an ERSAC-inspired ellipsoid converts covariance into a worst-case active contrast, conformal calibration sets a verifiable one-step radius, and shift diagnostics determine when abstention is mandatory. Under local exchangeability, the resulting no-trade gate gives a simultaneous one-step guarantee for the adaptively selected finite action.

Taken together, the earlier and public experiments sharpen the empirical message. In the previous proprietary full-OOS run, $\kappa = 0.20$ finishes above equal-weight variants in terminal NAV but below them in Sharpe; in the public Yahoo run, it finishes below them in both. The public policy holds the previous allocation on 66.1% of dates and substantially improves turnover, drawdown, and wealth relative to lower thresholds, yet active NAV remains below one against all four equal-weight baselines and active information ratios are near zero. The stable finding across the two long runs is therefore the importance of abstention, not a stable estimate of active alpha. The separate internal scenario diagnostic suggests that cross-asset state conditioning may be more consequential than the particular conditional generator, but it remains non-public and lacks formal uncertainty intervals.

The defensible deployment lesson is narrow: do not replace robust diversification with an unconstrained learned policy. Permit only bounded active overlays whose lower-confidence advantage clears costs and whose calibration regime remains credible. Demonstrating that the proposed conformal, ellipsoidal, cross-model, and shift-aware authorization layer can do so remains the central empirical task.

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This work was initiated during Tiantian Zhang’s internship at Boke Simu. The short rolling, previous full-OOS, conditional-scenario, and larger convex experiments used proprietary data accessed through Boke City / Boke Simu.

The public replication uses adjusted Yahoo Finance data retrieved through the repository’s documented workflow. The views expressed in this draft are those of the authors and do not necessarily represent Boke Simu, Boke City, Columbia University, or Tsinghua University.

Data, code, and disclaimer

The internal Wind/Boke data are proprietary and are not redistributed. The prior full-OOS tables and curves are retained from the authors’ earlier manuscript source; the revised source package includes machine-readable transcriptions and source hashes, but not the underlying proprietary panel or missing run logs. A public Yahoo Finance replication, including the data manifest, executed notebook, source code, action construction, experiment metadata, and archived summary outputs, is available on the [public replication branch](#) at experiment commit [f492551](#). The manuscript source package includes the archival public summary CSVs used in Tables 7–9, transcriptions of the retained prior experiments, and provenance records distinguishing the evidence tracks. Yahoo Finance data remain subject to the provider’s terms. This manuscript is for academic research only and is not investment advice.

A Optimization Baseline Definitions and Non-Comparable Internal Evidence

The following definitions make the intended cost convention explicit. Minimum variance is

$$w_t^{\text{MV}} \in \arg \min_{w \in \Delta^N} w^\top \Sigma_t w + c \|w - w_{t-1}\|_1. \quad (32)$$

A turnover-aware risk-budgeting formulation is

$$\tilde{w}_t^{\text{RP}} \in \arg \min_{w > 0} \frac{1}{2} w^\top \Sigma_t w - b \sum_{i=1}^N \log w_i + c \|w - w_{t-1}\|_1, \quad w_t^{\text{RP}} = \frac{\tilde{w}_t^{\text{RP}}}{\mathbf{1}^\top \tilde{w}_t^{\text{RP}}}. \quad (33)$$

A maximum-Sharpe reference can place cost in the return numerator,

$$w_t^{\text{MS}} \in \arg \max_{w \in \Delta^N} \frac{\hat{\mu}_t^\top w - r_f - c \|w - w_{t-1}\|_1}{\sqrt{w^\top \Sigma_t w}}. \quad (34)$$

For robust MVO, let $\rho_{t,i} = \sqrt{(\hat{\Sigma}_t^{\text{between}})_{ii}}$ be an explicit asset-level epistemic scale. One transparent specification is

$$\tilde{\mu}_{t,i} = \hat{\mu}_{t,i} - \tau_\mu \rho_{t,i}, \quad \tilde{\Sigma}_t = \Sigma_t + \hat{\Sigma}_t^{\text{between}}, \quad (35)$$

followed by

$$w_t^{\text{rob}} \in \arg \max_{w \in \Delta^N} \tilde{\mu}_t^\top w - \frac{\gamma}{2} w^\top \tilde{\Sigma}_t w - c \|w - w_{t-1}\|_1. \quad (36)$$

These specifications are proposed same-universe baselines; no seven-ETF numerical results are claimed.

Table 12: Internal Boke internship comparison retained from a larger equity/statistical-arbitrage setting. It is included only for provenance and is not comparable to the seven-ETF experiment.

Method	Final wealth	Main decision principle
IPMO-style path optimizer	1.5043	Multi-period return–risk–turnover path
Robust MVO	1.3980	Return shrinkage and covariance inflation
Hybrid 3D	1.2825	Robust signal, variance, connectivity penalty
3D frontier	1.2777	Mean return with PCA-factor concentration penalty
Alpha-weighted allocation	1.1978	Clipped signal weights with turnover smoothing

B Public Finite-Action Construction and Selection Counts

The public replication constructs ten candidate portfolios at every date and prepends the hold-previous-allocation action, giving 11 action IDs. Let $\sigma_{i,t}$ be the 20-day population standard deviation and let

$$m_{i,t} = \frac{\prod_{u=t-59}^t (1 + r_{i,u}) - 1}{\sigma_{i,t} + 10^{-6}} \quad (37)$$

be the 60-day risk-adjusted cumulative-momentum score. All nonnegative vectors are normalized to sum to one. During the first 120 observations, the ten non-hold candidates are set to equal weight.

Table 13: Complete public action-construction rules. Date-varying actions are determined entirely by lagged returns.

Action	Portfolio rule
Hold previous	Copy the portfolio held immediately before the decision.
Equal weight	Weight every ETF by $1/7$.
Inverse volatility	Normalize $1/(\sigma_{i,t} + 10^{-6})$.
Inverse variance	Normalize $1/(\sigma_{i,t} + 10^{-6})^2$.
Bond basket	Equal weight across AGG.P, LQD.P, and MUB.P; zero elsewhere.
Equity basket	Equal weight across VTI.P and IWM.P; zero elsewhere.
Gold	Weight GLD.P by one; zero elsewhere.
Commodity	Weight DBC.P by one; zero elsewhere.
Top-two momentum	Equal weight the two ETFs with the largest $m_{i,t}$.
Top-three momentum	Equal weight the three ETFs with the largest $m_{i,t}$.
Soft momentum	Normalize $\exp(\text{clip}(m_{i,t}, -5, 5) - \max_j \text{clip}(m_{j,t}, -5, 5))$.

Table 14: Archived action-selection counts over the 1,500 public OOS decisions.

κ	Hold	EW	InvVol	InvVar	Bond	Equity	Gold	Comm.	Top2	Top3	Soft
0.00	361	74	122	160	63	100	112	101	84	141	182
0.10	604	79	98	111	41	67	100	58	70	127	145
0.20	991	47	46	72	20	12	64	5	54	71	118

C Minimal Deployment Protocol

A point-in-time implementation of the proposed gate should execute the following steps on every roll:

1. Split the historical window into fit, gate-tuning, and conformal blocks with horizon-length purges; freeze feature normalization and kernel bandwidths using the fit block.
2. Fit a heterogeneous ensemble and generate action-value and one-step utility distributions for every finite action.
3. Compute within-model covariance, between-model covariance, and the cross-model kernel discrepancy.
4. Calibrate the utility ellipsoid radius with Equations (19)–(20) using only the purged calibration block.
5. Generate historical pseudo-deployment distributions for the four shift scores in Equation (28); freeze their quantile thresholds.
6. At deployment, propose an action with Equation (17); execute it only through Equation (29); log the lower bound, all shift scores, the accept/reject decision, realized counterfactual utilities, and rolling empirical coverage.

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